Year 11 Mathematics

Handling Data 4

4.1 Statistical Investigation and Collecting Data

Statistical Investigation and Collecting Data

STARTER · 5 MIN

Date: _____ **Title:** Statistical Investigation and Collecting Data

LEARNING OBJECTIVES

- Identify qualitative, discrete and continuous data
- Know primary and secondary data and choose a sample
- Write good questionnaire questions and spot bias

STARTER *Answer in your book. No calculators.*

1. Round 4.67 m to 1 d.p.
2. Work out 10% of 300
3. Work out $\frac{1}{4}$ of 120
4. Order 3.2, 3.05, 3.5 from smallest

Starter — Answers

STARTER

Tick or correct in green pen

1. **4.7 m**

2. $300 \div 10 = \mathbf{30}$

3. $120 \div 4 = \mathbf{30}$

4. **3.05**, **3.2**, **3.5**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can identify qualitative, discrete and continuous data	☐
SILVER	I know primary and secondary data and choose a sample	☐
GOLD	I can write good questionnaire questions and spot bias	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Types of Data

INTRODUCTION · 3 MIN

A statistical investigation starts with a question, then you collect the right data.

Qualitative

Described in words, not numbers.

Eye colour, favourite sport

Discrete

Counted: only exact values.

Number of siblings, shoe size

Continuous

Measured: any value in a range.

Height, time, mass

Primary or secondary

You collect it, or someone else did.

Survey vs internet data

Examples 1 and 2 — Types of Data

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Is favourite colour qualitative, discrete or continuous?

Example 2

Is the number of goals scored discrete or continuous?

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

It is described in words

Qualitative

Tip: words, not numbers.

Example 2

Goals are counted

Discrete

Tip: counted = discrete, measured = continuous.

Task 1

PRACTICE · 9 MIN

BRONZE *Write qualitative, discrete or continuous*

1. Hair colour
2. Number of pets
3. Time to run 100 m
4. Mass of a bag
5. Number of cars in a car park

★ Temperature at noon

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 **Qualitative**

2 **Discrete**

3 **Continuous**

4 **Continuous**

5 **Discrete**

★ **Continuous**

Examples 3 and 4 — Primary Data, Secondary Data and Samples

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Ali uses a website to find rainfall data. Is this primary or secondary data?

Example 4

A school has 800 students. Mia asks the first 20 students in the lunch queue. Why may the sample be biased?

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Someone else collected it

Secondary

Tip: primary = you collect it yourself.

Example 4

Only the queue is asked

Not a random sample

Tip: every student should have an equal chance.

Task 2

PRACTICE · 9 MIN

SILVER

Answer in your book

1. Measuring heights in your class: primary or secondary?

2. Using census data: primary or secondary?

3. Take a 10% sample of 450 students. How many?

4. Give one advantage of a sample over a census

5. Give one way to choose a random sample

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **Primary**

2. **Secondary**

3. $450 \div 10 = \mathbf{45}$

4. **Quicker and cheaper**

5. **Pick names from a hat**

Examples 5 and 6 — Questionnaires and Bias

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint number = $\frac{\text{group size}}{\text{total}}$ × sample size

Example 5

(2 marks)

Give two criticisms of this question: How much TV do you watch?

0–1 hours
1–2 hours
2–3 hours

Example 6

(2 marks)

Write a better question to find how much TV people watch.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

1 and 2 are in two boxes

No time period, no box above 3

Tip: look for overlaps, gaps and time frames.

Example 6

How many hours of TV
do you watch each week?

0–1 2–3 4–5 6 or more

Tip: include a time frame and boxes with no gaps.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint number = $\frac{\text{group size}}{\text{total}}$ × sample size

1. Give two criticisms of: 'Don't you agree school should start later?' Yes / No (2 marks)

2. Raj asks 10 people outside a gym how often they exercise. Give two reasons his results may be biased. (2 marks)

3. Write response boxes for: 'How many books did you read last month?' (2 marks)

4. A school has 360 girls and 240 boys. A sample of 50 is in the same proportion. How many girls? (3 marks)

5. Explain why asking only your friends is not a good sample. (1 mark)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. Leading question, **no unsure option**
2. Sample too small, **gym users exercise more**
3. **0 1–2 3–4 5 or more**
4. $360 \div 600 = 0.6$, $0.6 \times 50 =$ **30 girls**
5. Not random, **friends share views**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Is temperature discrete or continuous?

SILVER Data from a newspaper: primary or secondary?

GOLD Give one problem with: 'You like maths, don't you?'

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **Continuous**

SILVER **Secondary**

GOLD **Leading question**

Exam Practice — Types of Data

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(3 marks)

Write qualitative, discrete or continuous for each.

(a) the colour of a car (1)

.....

(b) the number of people in a car (1)

.....

(c) the time taken for a journey (1)

.....

Question 2

(2 marks)

Ben wants to find out how long students spend on homework.

(a) He asks the students in his class. Is this primary or secondary data? (1)

.....

(b) He also uses data from a website. Is this primary or secondary data? (1)

.....

Exam Practice — Samples

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

A school has 1200 students.

Sam wants to take a 5% sample.

(a) How many students should be in the sample? (1)

.....

(b) Give one way Sam could choose a random sample. (1)

.....

Question 4

(2 marks)

Lily wants to know how people travel to work.

She asks people at a train station at 8 am on a Monday.

Give two reasons why her sample may be biased.

.....

Exam Practice — Questionnaires and Stratified Samples

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint number = $\frac{\text{group size}}{\text{total}}$ × sample size

Question 5

(2 marks)

Here is a question from a survey.

How much do you spend on lunch?

☐ £0 – £2 ☐ £2 – £4 ☐ £5 or more

Give two criticisms of the response boxes.

.....

Question 6

(3 marks)

A club has 150 juniors and 90 seniors.

A sample of 40 members is taken in the same proportions.

How many juniors should be in the sample?

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) qualitative B1

(b) discrete B1

(c) continuous B1

Tip: counted: discrete. Measured: continuous.

Question 2

(a) primary B1

(b) secondary B1

Tip: primary data is collected by you.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 5% of 1200 = 60 B1

(b) number every student, B1

pick using random numbers

Tip: random = everyone has a chance.

Question 4

e.g. all at a station use the train B1

only one time and day B1

Tip: think about who is left out.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

£2 is in two boxes (overlap) — B1

no box for £4 to £5 (gap) — B1

Tip: boxes must not overlap or leave gaps.

Question 6

150 + 90 = 240 members — M1

$\frac{150}{240} \times 40$ — M1

= 25 juniors — A1

Tip: check: seniors = 15 and 25 + 15 = 40.